

UNIVERZA V LJUBLJANI
FAKULTETA ZA RAČUNALNIŠTVO IN INFORMATIKO

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**Ocenjevanje volatilnosti z
Markovskim prehajanjem na
finančnih trgih s sekvenčnimi Monte
Carlo metodami**

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MAGISTRSKI ŠTUDIJSKI PROGRAM DRUGE STOPNJE
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MENTOR: izr. prof. dr. Jure Demšar

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UNIVERSITY OF LJUBLJANA
FACULTY OF COMPUTER AND INFORMATION SCIENCE

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**Estimating Markov switching
volatility in financial markets using
sequential Monte Carlo methods**

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COMPUTER AND INFORMATION SCIENCE

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Worth mentioning in the acknowledgment is everyone who contributed to your thesis.

Vito Levstik, 2026

To all the flowers of this world.

*"The only reason for time is so that
everything doesn't happen at once."*

— Albert Einstein

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Abstract

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List of used acronyms

acronym	meaning
CA	classification accuracy
DBMS	database management system
SVM	support vector machine
...	...

Abstract

Title: Estimating Markov switching volatility in financial markets using sequential Monte Carlo methods

This sample document presents an approach to typesetting your BSc thesis using L^AT_EX. A proper abstract should contain around 100 words which makes this one way too short. A good abstract contains: (1) a short description of the tackled problem, (2) a short description of your approach to solving the problem, and (3) (the most successful) result or contribution in your thesis.

Keywords

computer, computer, computer

Povzetek

Naslov: Ocenjevanje volatilnosti z Markovskim prehajanjem na finančnih trgih s sekvenčnimi Monte Carlo metodami

V vzorcu je predstavljen postopek priprave magistrskega dela z uporabo okolja L^AT_EX. Vaš povzetek mora sicer vsebovati približno 100 besed, ta tukaj je odločno prekratek. Dober povzetek vključuje: (1) kratek opis obravnavanega problema, (2) kratek opis vašega pristopa za reševanje tega problema in (3) (najbolj uspešen) rezultat ali prispevek magistrske naloge.

Ključne besede

računalnik, računalnik, računalnik

Razširjeni povzetek

To je primer razširjenega povzetka v slovenščini, ki je obvezen za naloge pisane v angleščini. Razširjeni povzetek mora vsebovati vse glavne elemente dela napisanega v angleščini skupaj s kratkim uvodom in povzetkom glavnih elementov metode, glavnih eksperimentalnih rezultatov in glavnih ugotovitev. Razširjeni povzetek naj bo strukturiran v podpoglavja (spodaj je naveden le okvirni primer in je nezavezujoč). Čez palec navadno razširjeni povzetek nanese okoli 10 odstotkov obsega celotnega dela.

I Kratek pregled sorodnih del

II Predlagana metoda

III Eksperimentalna evaluacija

IV Sklep

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Chapter 1

Introduction

Chapter 2

Theoretical Background

We begin the theoretical background with the detailed description of the volatility, followed by the mathematical formulation of the state space models (SSM) and specifically the Markov switching stochastic volatility (MSSV) model. Then we present the theory of sequential Monte Carlo (SMC) methods, starting with bootstrap particle filter (BPF), auxiliary particle filter (APF), and then more advanced methods such as the particle Monte Carlo Markov Chain (PMCMC) and SMC².

2.1 Volatility

Volatility is a fundamental concept in financial econometrics and is commonly used as a measure of the variability or uncertainty of asset returns. In its most basic form, volatility refers to the standard deviation of returns over a given time horizon, and therefore quantifies the magnitude of fluctuations around the expected return.

Let P_t denote a price of an asset (stock, index, commodity, etc.) at time t . The log return of the asset is defined as

$$Y_t = \log(P_t) - \log(P_{t-1}) = \log\left(\frac{P_t}{P_{t-1}}\right). \quad (2.1)$$

The advantage of using log returns over simple returns is their additive prop-

erty over time and their ability to handle compounding effects [1]. Throughout this thesis, when we refer to returns, we mean log returns.

While returns are directly observable, volatility is a characteristic of the underlying return distribution and is not directly observable from a single realization. At each time point t , only one return Y_t is observed, whereas volatility is defined in terms of the variability of the entire distribution generating the returns. As a consequence, the time-varying volatility of financial returns must be inferred from data rather than directly measured.

To formalize this idea, let $\mathcal{F}_{t-1}$ denote the information available up to time $t - 1$. The conditional mean and variance of the return are defined as

$$\mu_t = \mathbb{E}[Y_t | \mathcal{F}_{t-1}], \quad (2.2)$$

$$\sigma_t^2 = \text{Var}(Y_t | \mathcal{F}_{t-1}) = \mathbb{E}[(Y_t - \mu_t)^2 | \mathcal{F}_{t-1}]. \quad (2.3)$$

The quantity σ_t is referred to as the conditional volatility. In many financial applications, the conditional mean is assumed to be negligible relative to the variability of returns, so that the focus is primarily on modelling the conditional volatility process.

When high-frequency intraday data are available, volatility can be estimated nonparametrically using the concept of realized volatility. Suppose that within a fixed time interval (e.g., one day), we observe intraday log returns $r_{t,i}$ for $i = 1, \dots, m$. The realized variance is defined as

$$RV_t = \sum_{i=1}^m r_{t,i}^2, \quad (2.4)$$

Under certain regularity conditions, as the sampling frequency increases (i.e., $m \rightarrow \infty$), the realized variance converges to the quadratic variation of the log-price process, which in the absence of jumps corresponds to the integrated variance over the interval. This provides a theoretical justification for using realized volatility as a consistent estimator of latent daily volatility.

2.2 State Space Models

A state space model describes a dynamical system in which an unobserved (latent) stochastic process drives the evolution of an observed process. The key idea is that the system evolves over time through hidden states, while only noisy measurements of these states are available.

Let $\{X_t\}_{t \geq 0}$ be a collection of random variables (also called a stochastic process) defined on abstract probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with values in a measurable space $(\mathcal{X}, \mathcal{B}(\mathcal{X}))$. Similarly, let $\{Y_t\}_{t \geq 1}$ be another stochastic process defined on the same probability space taking values in $(\mathcal{Y}, \mathcal{B}(\mathcal{Y}))$. Together, $\{(X_t, Y_t)\}_{t \geq 1}$ forms a stochastic process taking values in the joint space $(\mathcal{X} \times \mathcal{Y}, \mathcal{B}(\mathcal{X}) \otimes \mathcal{B}(\mathcal{Y}))$ [2].

The process $\{X_t\}_{t \geq 0}$ is *hidden* (or *latent*) from the observer and modelled as a *Markov process*, while $\{Y_t\}_{t \geq 1}$ can be *observed* and is conditionally independent given the state process $\{X_t\}_{t \geq 0}$. We model the system via following densities:

$$X_0 \sim \chi(x_0|\theta), \quad (2.5)$$

$$X_t|X_{t-1} = x_{t-1} \sim p(x_t|x_{t-1}, \theta), \quad (2.6)$$

$$Y_t|X_t = x_t \sim q(y_t|x_t, \theta), \quad (2.7)$$

where χ is the *initial density*, p is the *transition density*, and q is the *observation* or *emission density*. With $\theta \in \Theta$ we denote the vector of model parameters [2, 3, 4].

A potential problem with this definition is that conditional distribution of X_t given X_{t-1} may not always admit a density. A proper way would be to define the state space model in a more measure theoretic way using probability kernels [2]. However, we will not deal with this more general definition in this thesis, as our models will have nice densities.

2.3 MSSV Model

The Markov switching stochastic volatility (MSSV) model is a state space model where the hidden state X_t is a tuple (H_t, S_t) , where H_t is the *log volatility* and S_t is a discrete *regime indicator*. This means that $\{H_t\}_{t \geq 0}$ is a stochastic process taking values in $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ and $\{S_t\}_{t \geq 0}$ is a stochastic process taking values in $(\{1, 2, \dots, K\}, 2^{\{1, 2, \dots, K\}})$, where K is the number of regimes. The latent state space is then simply $\mathcal{X} = \mathbb{R} \times \{1, 2, \dots, K\}$, equipped with the product σ -algebra $\mathcal{B}(\mathbb{R}) \otimes 2^{\{1, 2, \dots, K\}}$.

The observation density is defined as

$$Y_t | H_t = h_t, S_t = s_t \sim \mathcal{N}(0, \exp(h_t)), \quad (2.8)$$

where $\mathcal{N}(\mu, \sigma^2)$ denotes the normal distribution with mean μ and variance σ^2 . The transition density is defined as

$$H_t | H_{t-1} = h_{t-1}, S_t = s_t \sim \mathcal{N}(\mu_{s_t} + \phi(h_{t-1} - \mu_{s_t}), \eta^2), \quad (2.9)$$

$$S_t | S_{t-1} = s_{t-1} \sim \text{Categorical}(P_{s_{t-1}, :}), \quad (2.10)$$

where μ_{s_t} are regime-specific means, ϕ is the persistence parameter, η is the standard deviation of the log volatility process and P is the $K \times K$ regime transition matrix defined as

$$P = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1K} \\ p_{21} & p_{22} & \cdots & p_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ p_{K1} & p_{K2} & \cdots & p_{KK} \end{bmatrix},$$

where $p_{ij} = \mathbb{P}(S_t = j | S_{t-1} = i)$ is the probability of transitioning from regime i to regime j . Note that the rows of P must sum to 1, i.e., $\sum_{j=1}^K p_{ij} = 1$ for all i . Observe that the log volatility process is essentially an AR(1) process with regime-specific means [5].

The description of the MSSV model is complete with the specification of

the initial (prior) density. We define the initial density as

$$H_0|S_0 = s_0 \sim \mathcal{N}\left(\mu_{s_0}, \frac{\eta^2}{(1-\phi^2)}\right), \quad (2.11)$$

$$S_0 \sim \text{Categorical}\left(\frac{1}{K}, \frac{1}{K}, \dots, \frac{1}{K}\right). \quad (2.12)$$

The choice of H_0 is motivated by the stationary distribution of the AR(1) process, which has mean μ_{s_0} and variance $\frac{\eta^2}{(1-\phi^2)}$, whereas the choice of S_0 is motivated by the lack of prior knowledge about the initial regime, so we assume that all regimes are equally likely.

The following assumptions are made [5]:

1. The persistence ϕ is in range $(-1, 1)$ to ensure stationarity of the AR(1) process.
2. Each regime should correspond to at least one time point.
3. The regime means are ordered such that $\mu_1 < \mu_2 < \dots < \mu_K$ to avoid label switching issues. We do this with the following reparameterization:

$$\mu_i = \mu_{i-1} + \exp(\delta_i), \quad i = 2, \dots, K,$$

where δ_i is a real valued parameter.

The parameters of the MSSV model θ are then given by

$$\theta = \{\mu_1, \delta_2, \dots, \delta_K, \phi, \eta, P\},$$

but due to requirement of matrix P being row stochastic, we actually have $K(K-1)$ free parameters in P . Therefore, the total number of free parameters is $K + 2 + K(K-1) = K^2 + 2$.

Note that in case of $K = 1$, the MSSV model reduces to the standard stochastic volatility (SV) model, in which only the log volatility H_t is a latent variable and there are no regimes.

Chapter 3

Methodology

Chapter 4

Results

Chapter 5

Discussion

Chapter 6

Conclusion

Appendix A

Title of the appendix 1

Example of the appendix.

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